



## Lab 2

### Arduino World

**Course:** RoboLink: Build and Drive Your Smartphone-Controlled Robot

**Program:** Mini Course Program 2026, Carleton University

**Instructors:** Dr. Mahmoud Sayed and Mahmoud Yassin

#### Lab Mission

Today you will enter Arduino World. You will learn how to open the Arduino IDE, upload your first program, and use simple electronic parts to control light and sound. By the end of the day, a push button will control LED blinking speeds, LED colours, and a buzzer.

Group Member Names \_\_\_\_\_

Date \_\_\_\_\_

## 1. Learning Goals

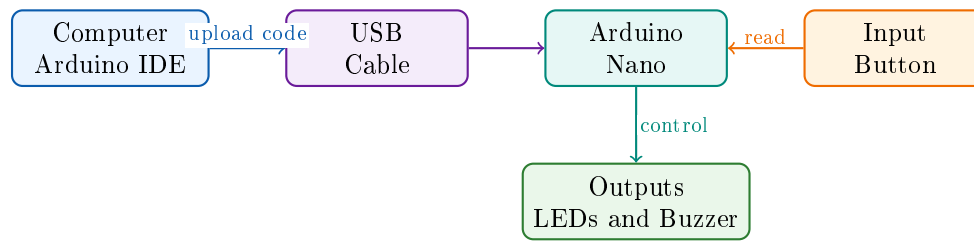
By the end of this lab, you should be able to:

- Open the Arduino IDE and describe what it is used for.
- Select the correct Arduino Nano board, processor, and port.
- Upload the built-in Blink example to the Arduino.
- Explain the difference between `setup()` and `loop()`.
- Wire LEDs, a push button, and a buzzer safely on a breadboard.
- Use a push button to change the blinking speed of an LED.
- Use one push button to choose between multiple LED colours and buzzer speeds.

## 2. Big Idea

#### An Arduino program connects code to the real world.

The computer sends a program to the Arduino. The Arduino then follows that program again and again. It can read inputs, such as a button, and control outputs, such as LEDs and buzzers.



### 3. Materials

|                                    |                                                           |
|------------------------------------|-----------------------------------------------------------|
| <b>Arduino Nano ATmega168</b>      | Runs the code and controls the circuit.                   |
| <b>Computer with Arduino IDE</b>   | Used to write, open, verify, and upload Arduino sketches. |
| <b>USB cable</b>                   | Connects the Arduino Nano to the computer.                |
| <b>Breadboard and jumper wires</b> | Used to build the circuit without soldering.              |
| <b>LEDs and resistors</b>          | LEDs create light. Resistors protect the LEDs.            |
| <b>Push button</b>                 | Gives the Arduino an input signal.                        |
| <b>Three-pin buzzer module</b>     | Creates sound in Experiment 3.                            |

### 4. Safety and Setup Notes

#### Check Wiring Before Power

Ask your teacher to check the circuit before plugging in the Arduino. A wrong connection can stop the circuit from working or can damage a part.

#### Use Resistors With LEDs

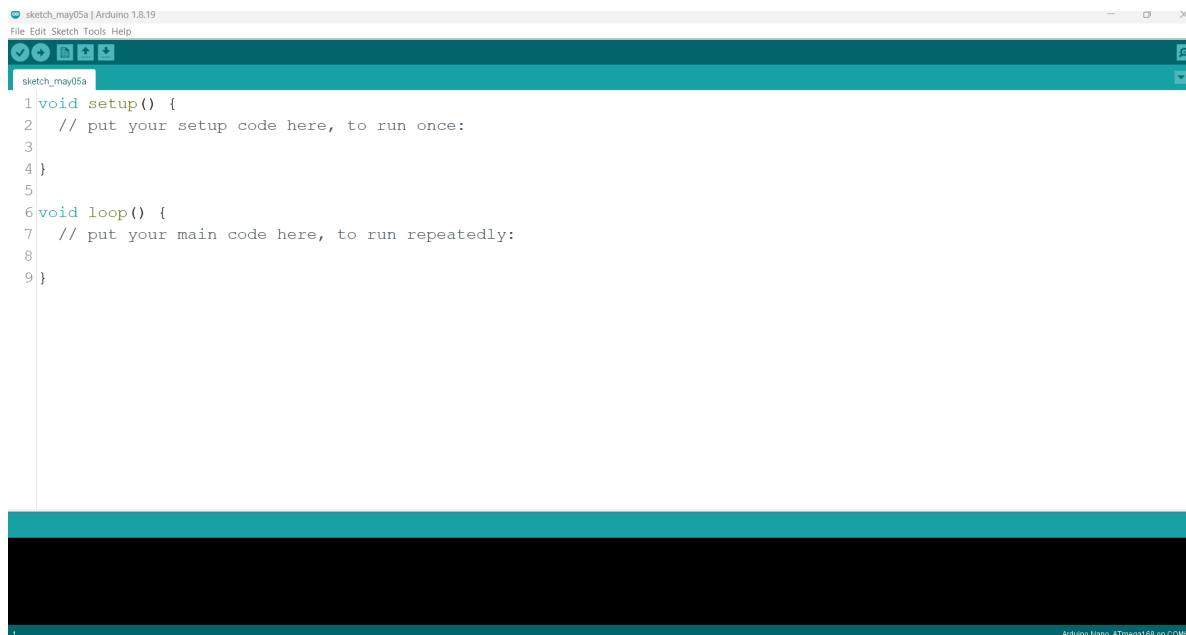
Each external LED should have a resistor in series with it. A common value is 220 ohms to 330 ohms.

#### Button Logic Today

The push button circuits use `INPUT_PULLUP`. This means the Arduino reads **HIGH** when the button is not pressed and **LOW** when the button is pressed.

### 5. Meet the Arduino IDE

The Arduino IDE is the program on your computer that lets you write and upload Arduino code. Arduino programs are called **sketches**.

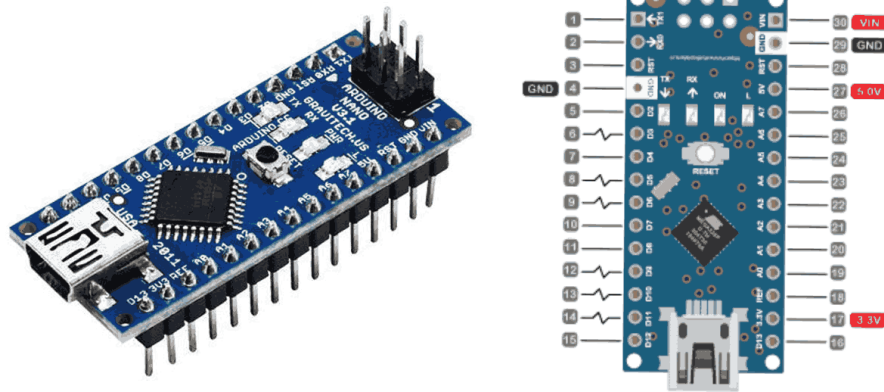


**Arduino IDE.** The IDE is used to open code, verify code, upload code, and read messages from the Arduino.

| IDE Part              | What It Does                                                |
|-----------------------|-------------------------------------------------------------|
| <b>Verify button</b>  | Checks whether the sketch has code errors before uploading. |
| <b>Upload button</b>  | Sends the sketch from the computer to the Arduino Nano.     |
| <b>Sketch area</b>    | The place where the Arduino code appears.                   |
| <b>Message area</b>   | Shows errors, upload messages, and helpful information.     |
| <b>Serial Monitor</b> | Lets the Arduino print messages back to the computer.       |

## First-Time Board Setup

1. Open the Arduino IDE.
2. Plug the Arduino Nano into the computer using the USB cable.
3. Select **Tools -> Board -> Arduino AVR Boards -> Arduino Nano**.
4. Select **Tools -> Processor -> ATmega168**.
5. Select the correct **Tools -> Port**. It usually looks like a COM port on Windows.
6. Click **Verify** to check the code.
7. Click **Upload** to send the code to the Arduino.



**Arduino Nano pinout.** Pin names such as D2, D8, D9, and D13 are used in today's circuits.

**Teacher checkpoint:** Show your teacher that your Arduino Nano board, ATmega168 processor, and port are selected correctly.

## 6. Arduino Sketch Structure

Most Arduino sketches have two main parts:

```

1 void setup() {
2   // Runs once when the Arduino starts.
3 }
4
5 void loop() {
6   // Runs again and again while the Arduino has power.
7 }
```

|                             |                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------|
| <code>setup()</code>        | Use this for settings that happen once, such as choosing which pins are outputs. |
| <code>loop()</code>         | Use this for actions that repeat, such as blinking an LED or checking a button.  |
| <code>pinMode()</code>      | Tells the Arduino whether a pin is an input or an output.                        |
| <code>digitalWrite()</code> | Turns a digital output pin HIGH or LOW.                                          |
| <code>digitalRead()</code>  | Reads whether a digital input pin is HIGH or LOW.                                |

## 7. Download the Day 2 Code

### Code Link for Experiment 2 and Experiment 3

Download the Day 2 Arduino code from:

[https://github.com/MYMahmoudYassin/MPC-2026-RoboLink/tree/main/Day2\\_Arduino\\_Basics](https://github.com/MYMahmoudYassin/MPC-2026-RoboLink/tree/main/Day2_Arduino_Basics)

Open the correct folder in the Arduino IDE before uploading:

- Button\_controlled\_LED\_blinking\_speed for Experiment 2.
- Control\_the\_Blinking\_of\_an\_LED\_and\_Buzzer for Experiment 3.

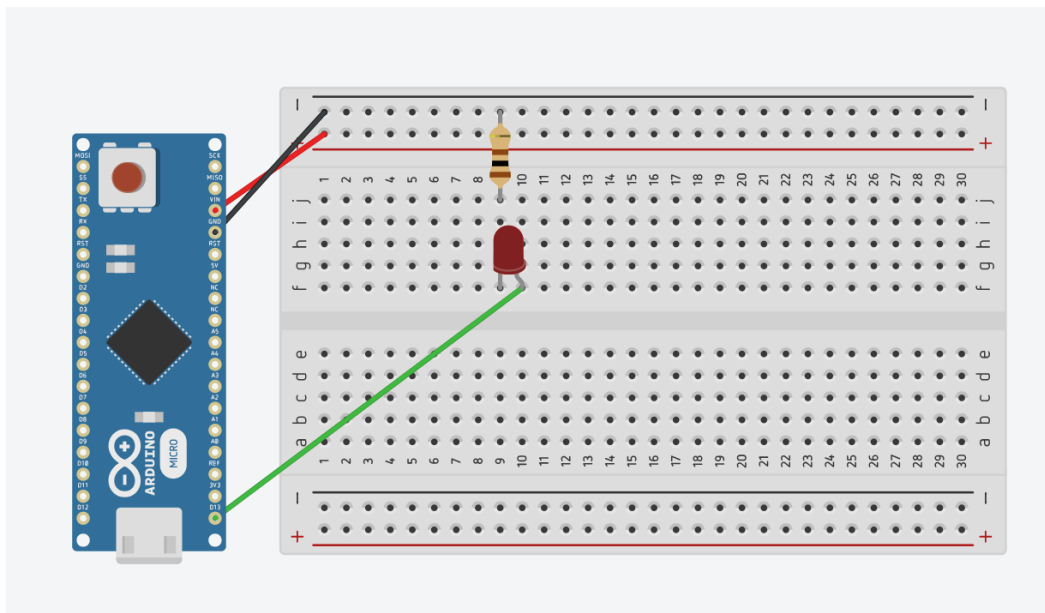
## 8. Experiment 1: Built-In Blink Example

### Goal

Upload your first Arduino program using an example that is already included in the Arduino IDE.

### Circuit

For the first test, you can use the built-in LED on pin 13. If your teacher asks you to use an external LED, connect it as shown below.



**Blink circuit.** An external LED can be connected to D13 with a resistor.

### Procedure

1. Open the Arduino IDE.

2. Go to **File -> Examples -> 01.Basics -> Blink**.
3. Check that the board is **Arduino Nano** and the processor is **ATmega168**.
4. Click **Upload**.
5. Watch the LED turn on and off.
6. Change both `delay(1000)` lines to `delay(500)`.
7. Upload again and observe what changed.

### Key Code Idea

```
1 digitalWrite(LED_BUILTIN, HIGH); // LED on
2 delay(1000);                     // wait 1 second
3 digitalWrite(LED_BUILTIN, LOW);  // LED off
4 delay(1000);                     // wait 1 second
```

| Delay Value | Prediction | Observation |
|-------------|------------|-------------|
| 1000 ms     |            |             |
| 500 ms      |            |             |
| 200 ms      |            |             |

## 9. Experiment 2: Push Button LED Blinking Speed

### Goal

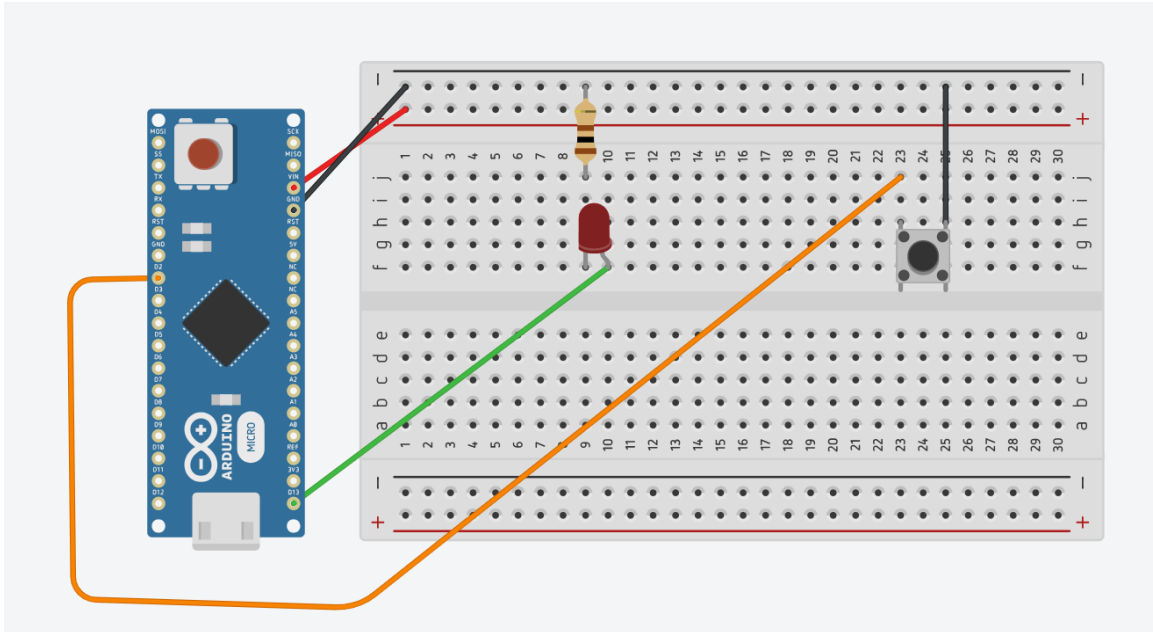
Use one push button to change the blinking speed of one LED.

### Open the Code

Open this sketch:

Button\_controlled\_LED\_blinking\_speed/Button\_controlled\_LED\_blinking\_speed.ino

## Circuit



**Experiment 2 circuit.** The button changes the LED blinking speed.

| Part             | Arduino Pin | Connection Notes                        |
|------------------|-------------|-----------------------------------------|
| LED positive leg | D13         | Use a resistor with an external LED.    |
| LED negative leg | GND         | Connect to ground.                      |
| Button side 1    | D2          | The code uses INPUT_PULLUP.             |
| Button side 2    | GND         | Pressing the button connects D2 to GND. |

## What the Code Uses

```

1 const byte LED_PIN = 13;
2 const byte BUTTON_PIN = 2;
3
4 const unsigned long BLINK_TIMES[] = {
5     1000, 700, 400, 200, 100
6 };

```

| Speed | Blink Time | What You Should See |
|-------|------------|---------------------|
| 1     | 1000 ms    | Slow blink          |
| 2     | 700 ms     | A little faster     |
| 3     | 400 ms     | Medium speed        |
| 4     | 200 ms     | Fast                |
| 5     | 100 ms     | Fastest             |

## Procedure

1. Build the circuit.
2. Ask your teacher to check the wiring.
3. Open the Experiment 2 sketch.

4. Click **Verify**. If there are no errors, click **Upload**.
5. Press the button once. Record what happens.
6. Keep pressing the button until the speed returns to slow again.

**Why not use only `delay()`?**

The Experiment 2 code uses `millis()` so the Arduino can keep checking the button while the LED is blinking. This is more flexible than stopping the whole program with long `delay()` commands.

**Observation:** Which speed was easiest to count with your eyes?

## 10. Experiment 3: Multiple LED Colours and Buzzer

### Goal

Use one button to choose between five LED colours. The selected LED blinks, and the buzzer beeps at the same speed.

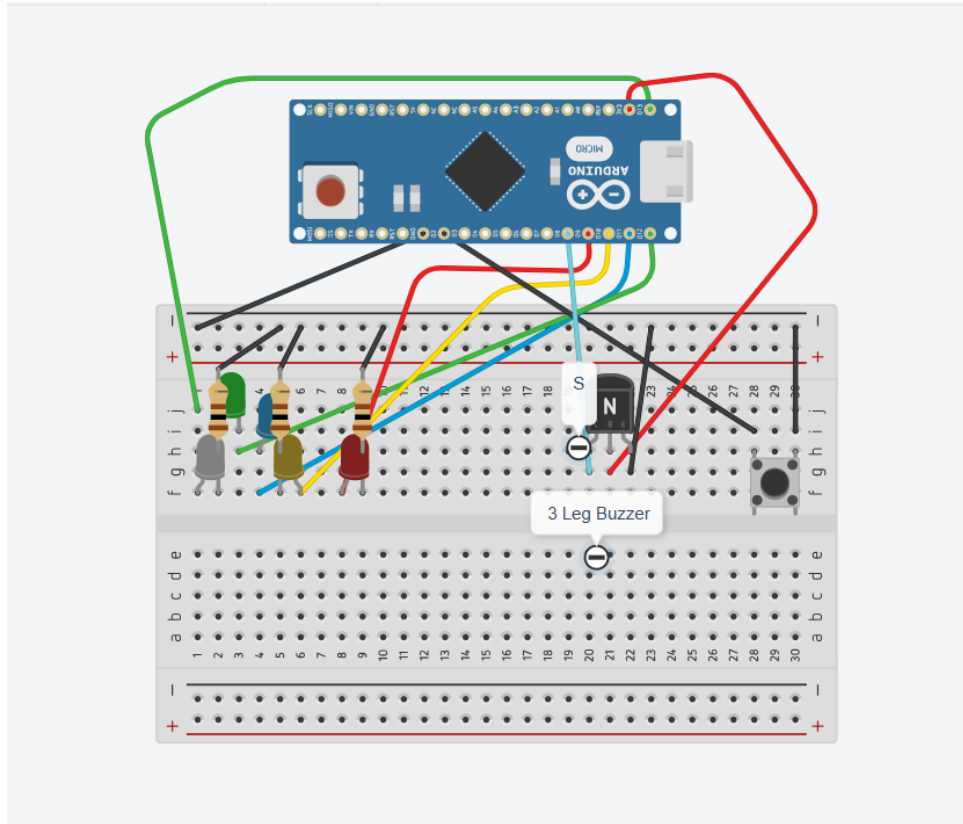
### Open the Code

Open this sketch:

`Control_the_Blinking_of_an_LED_and_Buzzer/Control_the_Blinking_of_an_LED_and_Buzzer.ino`



## Circuit



**Experiment 3 circuit.** The selected LED blinks and the buzzer beeps.



**Three-pin buzzer module.** Connect signal to D8, VCC to 5V, and GND to GND.

| Part               | Arduino Pin | Connection Notes                               |
|--------------------|-------------|------------------------------------------------|
| White LED          | D13         | Speed 1, slowest. Use a resistor.              |
| Green LED          | D12         | Speed 2. Use a resistor.                       |
| Blue LED           | D11         | Speed 3. Use a resistor.                       |
| Yellow LED         | D10         | Speed 4. Use a resistor.                       |
| Red LED            | D9          | Speed 5, fastest. Use a resistor.              |
| Button             | D2 and GND  | Pressing the button selects the next speed.    |
| Buzzer signal      | D8          | The code uses <code>tone()</code> on this pin. |
| Buzzer VCC and GND | 5V and GND  | Match the labels on your buzzer module.        |

## What the Code Uses

```

1 const byte BUTTON_PIN = 2;
2 const byte BUZZER_PIN = 8;
3
4 const byte LED_PINS[] = {
5     13, 12, 11, 10, 9
6 };
7
8 const unsigned int BUZZER_FREQUENCY = 1000;

```

| Speed | LED Colour | Blink Time | Expected Output          |
|-------|------------|------------|--------------------------|
| 1     | White      | 1000 ms    | Slow blink and slow beep |
| 2     | Green      | 700 ms     | A little faster          |
| 3     | Blue       | 400 ms     | Medium blink and beep    |
| 4     | Yellow     | 200 ms     | Fast blink and beep      |
| 5     | Red        | 100 ms     | Fastest blink and beep   |

## Procedure

1. Build the circuit using the wiring table.
2. Ask your teacher to check the LED directions, resistors, button, and buzzer.
3. Open the Experiment 3 sketch.
4. Click **Verify**. If there are no errors, click **Upload**.
5. Press the button once and record which LED turns on.
6. Keep pressing the button until the red LED is selected.
7. Press the button again and check that the program returns to the white LED.

### Important Code Pattern

The code stores the LED pins in an array. This lets one number, `currentSpeedNumber`, choose the matching LED, blink time, and buzzer speed pattern.

**Observation:** Which LED colour matched the fastest speed? How did you know?

## 11. Troubleshooting

| Problem                                | Possible Cause                                                | What To Try                                           |
|----------------------------------------|---------------------------------------------------------------|-------------------------------------------------------|
| <b>Upload fails</b>                    | Wrong board, processor, or port                               | Select Arduino Nano, ATmega168, and the correct port. |
| <b>No port appears</b>                 | USB cable or driver problem                                   | Try another USB cable or ask your teacher for help.   |
| <b>LED does not turn on</b>            | LED direction is reversed, missing resistor, or wrong pin     | Check the long leg, resistor, D13, and GND.           |
| <b>Button does nothing</b>             | Button is in the wrong breadboard row or not connected to GND | Check D2 and GND wiring.                              |
| <b>Button changes speed many times</b> | Button bounce is being detected                               | Make sure the debounce code was not removed.          |
| <b>Buzzer does not beep</b>            | Signal, VCC, or GND is wrong                                  | Check D8, 5V, and GND on the buzzer module.           |
| <b>Wrong colour turns on</b>           | LED order is different from the code                          | Check D13, D12, D11, D10, and D9.                     |

## 12. Reflection Questions

Answer in full sentences.

1. What is the job of the Arduino IDE?
2. What happens inside `setup()`?
3. Why does `loop()` run again and again?
4. Why does a button connected with `INPUT_PULLUP` read `LOW` when pressed?
5. Why is `millis()` useful for the push button blinking experiments?
6. In Experiment 3, how does the code know which LED colour to blink?
7. What is one change you would make to improve Experiment 3?

## 13. Exit Ticket

**In one or two sentences:** Explain how a push button can change what an Arduino does.

## 14. Teacher Check

### Teacher Checklist

|                                                                          |              |
|--------------------------------------------------------------------------|--------------|
| <input type="checkbox"/> Arduino IDE opened                              | Notes: _____ |
| <input type="checkbox"/> Arduino Nano and ATmega168 selected             | Notes: _____ |
| <input type="checkbox"/> Experiment 1 Blink uploaded                     | Notes: _____ |
| <input type="checkbox"/> Experiment 2 circuit checked                    | Notes: _____ |
| <input type="checkbox"/> Experiment 2 code uploaded and tested           | Notes: _____ |
| <input type="checkbox"/> Experiment 3 circuit checked                    | Notes: _____ |
| <input type="checkbox"/> Experiment 3 code uploaded and tested           | Notes: _____ |
| <input type="checkbox"/> Students completed observations and reflections | Notes: _____ |